Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/29/2008
PI Name:	Turner, Ronald E Ph.D.		
Project Title:	Solar Particle Event Observations with MARIE, and Correlations with Earth and Near-Earth Instruments		
Division Name:	Human Research		
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Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Zip Code:	22206	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
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No. of PhD Candidates:		**No. of Master' Degrees:**	
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Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: received NCE to 7/31/2006 per J. Dardano/JSC (5/06)		
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Task Description:	The Odyssey mission presents an exciting opportunity for advances in radiation risk management of a future human mission to Mars. A major objective of this mission is to characterize the Martian radiation environment to support future human missions to Mars. In addition, the MARIE instrument, designed to measure the energetic particle flux at Mars, can be used to provide multipoint observations of solar particle events (SPEs) in the critical region of the heliosphere needed to reduce the in-flight radiation risk to a future Mars-bound crew. MARIE collected data in Mars orbit from March 2002 to the end of October 2003. While efforts continue to revive the instrument, the data collected to date have already been very valuable and offer insight into the heliospheric evolution and structure of solar particle events. This work will examine MARIE data collected during SPEs, extract critical information including timing and magnitude of event onset, peak, possible shock-enhanced peaks, and subsequent decay.		

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Initial work examined MARIE data collected during SPEs, extracted critical information including timing and magnitude of event onset, peak, possible shock-enhanced peaks, and subsequent decay. The data can be correlated with Earth and Near-Earth particle flux data (from GOES, ACE or other spacecraft) on the same event. The data can also be correlated with images from SOHO and other solar observatories of the solar source event. Because the separation of Earth and Mars varies over the nearly two-year orbital period of Mars, Odyssey observations help establish the heliospheric extent of SPEs (when do both Earth and Mars experience the same SPE, and when are SPEs noted at one, but not the other). These data can be used to impose constraints on SPE models, even when the observational information is limited to event onset, peak, and decay times.
Relationship to Odyssey objectives

While the major focus of the Odyssey mission is on Mars itself, the MARIE instrument was the result of collaboration between the NASA Space Science Enterprise and the Human Exploration and Development of Space Enterprise, (now the Science Directorate and the Space Exploration Directorate). MARIE was one of the first instruments flown on a Mars mission specifically designed to address issues that will impact future human exploration of the planet. There is now a close working relationship between the planetary science community and the human exploration community that benefits both. One example of that working relationship is the representation of the Mars Exploration Program/Payload Analysis Group (MEPAG), which maintains a living document to identify the objectives, investigations, and measurements needed for the exploration of Mars (See Mars Science Goals, Objectives, Investigations, and Priorities: 2005, Aug 5, 2005, http://mepag.jpl.nasa.gov/)

Having a sensitive radiation sensor at Mars provides a unique opportunity to further our understanding of SPEs through the collection of distributed, multipoint observations. From March 2002 through October 2003 (the period during which MARIE was active) the Earth/Mars heliolongitude separation varies from zero to a full +/- 180 degrees. Heliolatitude separation varies from –46 to over 38 degrees. During the extended mission period, mid 2005 to mid 2007, the angle again varies from zero to a full +/- 180 degrees, while the Heliolatitude variation varies between +/- 45 degrees. This variation provides a natural laboratory to probe various geometries of interest. CMEs expand outward radially from the sun. They vary in extent from 10s of degrees to well over 180 degrees (Hundhausen, 1997). The faster CMEs responsible for SPEs are generally the larger ones in extent. When Earth and Mars are within 90 degrees of one another, they are likely to both be affected by any shock wave associated with these larger CMEs. The extent, strength, and evolution of this shock enhanced peak, at the nose of the shock and along the leading and trailing edges of the shock, provides insight into the physics of the acceleration process and is of vital interest to astronaut safety (Reames, 1999)

The character of the SPE relative to the location of the center of the CME is one of the most important parameters for forecasting the evolution of the first few hours of the SPE (away from the shock enhanced peak). The accelerated particles that have escaped from the shock move out through the heliosphere “clockwise” along the “Archimedean spiral” of the interplanetary magnetic field line. The curvature of the spiral is a result of the Sun’s rotation and it depends on the solar wind speed. The solar wind typically varies from 300 to 500 km/sec. The ambient solar wind speed can be measured near Earth and estimated for Mars. The faster the wind, the lower the curvature.

Solar Particle Events are roughly correlated with the solar cycle, occurring more frequently during seven years of solar active years than the three or so years near solar minimum (Feynman, et al., 1993). The current solar cycle (Cycle 23) has produced several large solar particle events, many during the period MARIE was collecting data. Solar activity and SPEs decrease during Solar Minimum, and the period of the PS will overlap Solar Minimum. However, during the approach to Solar Minimum, there have been several significant SPEs and a few periods of minor events. Events during Solar Minimum are generally isolated events occurring in a simple solar wind background. These events are particularly good for testing models of event propagation. However, this requires multipoint observations. Continuing observations at Mars with the same data source used since 2002 (the GRS data) would continue to complement data collected at Earth by GOES and other spacecraft.

The MARIE instrument

The MARIE instrument was optimized for high energy, low count rate GCR and secondary particles (including neutrons scattered up from the planetary atmosphere). The very high count rate experienced during large SPEs saturated the instrument. The PI prepared techniques to collect critical data as much as possible during SPEs as part of the core MARIE mission.

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References

Feynman, J., G. Spitale, and J. Wang, Interplanetary proton fluence model: JPL 1991, J. Geophys. Res, 98, A8, pp. 13,281–13,294, August 1993.

Hundhausen, A., Introduction, Coronal Mass Ejections — Geophysical Monograph 99, American Geophysical Union, Washington D.C., 1997.

Reames, D., “Solar energetic particles: Is there time to hide?,” Radiation Measurements, Vol 30, Issue 3, pp. 297-308, November 1999.

Fourth Year Publications

• Abstract and Poster, “MARIE Observations of Solar Particle Events” at the 2005 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2005.

	• Invited talk, Wintergreen Conference Space Weather and the Vision for Space Exploration, “Space Weather Challenges Intrinsic to the Human Exploration of Space,” Charlottesville, Virginia October 2005. • Invited talk, Wintergreen Conference on Space Weather Contributions to the Space Exploration Initiative, “Risk Management Strategies During Solar Particle Events on Human Missions to the Moon and Mars,” Wintergreen, Virginia, October 2005. • Invited talk, Space Weather Week, “Radiation Risk Management on Human Missions to the Moon and Mars,” Boulder, Colorado, April 2006 • Poster, “Solar Energetic Particles in Near-Mars Space” J.G. Luhmann, D.A. Brain, C. Lee, G.T. Delory, R. Turner, C.J. Zeitlin, J.G. Lyon, EGU, April, 2006
Bibliography Type:	Description: (Last Updated: 01/14/2008)
Abstracts for Journals and Proceedings	Turner R. "MARIE Observations of Solar Particle Events." 2005 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2005. 2005 Bioastronautics Investigators' Workshop, Galveston, Texas, January 2005. , Jan-2005
Abstracts for Journals and Proceedings	Luhmann JG, Brain DA, Lee C, Delory GT, Turner R, Zeitlin CJ, Lyon JG. "Solar Energetic Particles in Near-Mars Space." EGU General Assembly, Vienna, Austria, April, 2006. EGU General Assembly meeting, April, 2006. , Apr-2006
Books/Book Chapters	Turner R. "Space Weather Challenges Intrinsic to the Human Exploration of Space." in "Solar eruptions and energetic particles." Ed. N. Gopalswamy, R. Mewaldt, and J. Torsti. Washington, DC : American Geophysical Union, c2006. (Geophysical monograph series ; 165), Jul-2006
Papers from Meeting Proceedings	Turner R. "Space Weather Challenges Intrinsic to the Human Exploration of Space. Invited talk." Wintergreen Conference Space Weather and the Vision for Space Exploration, Charlottesville, Virginia, October 2005. Wintergreen Conference Space Weather and the Vision for Space Exploration, Charlottesville, Virginia, October 2005. , Oct-2005
Papers from Meeting Proceedings	Turner R. "Risk Management Strategies During Solar Particle Events on Human Missions to the Moon and Mars. Invited talk." Wintergreen Conference on Space Weather Contributions to the Space Exploration Initiative, Wintergreen, Virginia, October 2005. Wintergreen Conference on Space Weather Contributions to the Space Exploration Initiative, Wintergreen, Virginia, October 2005. , Oct-2005
Papers from Meeting Proceedings	Turner R. "Radiation Risk Management on Human Missions to the Moon and Mars. Invited talk." Space Weather Week, Boulder, Colorado, April 2006. Space Weather Week, Boulder, Colorado, April 2006. , Apr-2006